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Two-timepoint multidomain follow-up of post-COVID condition and ME/CFS: overlapping autonomic, small-fiber, and cognitive changes

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Abstract

Background Post-COVID condition (PCC) and Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS) show marked clinical overlap, suggesting a shared post-infectious pathophysiology. This study aims to characterize the longitudinal change of autonomic function, small-fiber integrity, cognitive performance, and clinical symptoms in PCC and ME/CFS, and to determine whether trajectories differ between diagnostic groups.

Methods Thirty-eight participants (21 PCC, 17 ME/CFS) underwent two standardized evaluations separated by a median of 31 months. Assessments included comprehensive autonomic testing, small-fiber evaluation, and an extensive neuropsychological battery.

Results ME/CFS showed longer disease duration than PCC at baseline (median 42 vs. 12 months), while the interval between evaluations was comparable (31 vs. 30 months). Baseline profiles were largely overlapping, although ME/CFS showed nominally higher QST warm detection thresholds ($p=0.034$), greater autonomic symptom burden ($p=0.038$), and lower hemodynamic scores ($p=0.019$), none surviving FDR correction. Cross-domain analyses linked small-fiber symptoms with autonomic symptom burden ($\text{Rho}=0.65$, $\text{pFDR}=0.002$) and fatigue ($\text{Rho}=0.55$, $\text{pFDR}=0.018$), while fatigue was negatively associated with processing speed ($\text{Rho}=-0.57$, $\text{pFDR}=0.004$), attention ($\text{Rho}=-0.49$, $\text{pFDR}=0.018$), and executive function ($\text{Rho}=-0.44$, $\text{pFDR}=0.047$). Rank-transformed mixed-effects models identified FDR-corrected Time effects, with increases in CHEPs ($\text{pFDR}<0.001$) and verbal memory ($\text{pFDR}=0.010$), and decreases in processing speed ($\text{pFDR}=0.006$) and QST cold thresholds ($\text{pFDR}=0.038$).

Conclusions PCC and ME/CFS showed broadly overlapping multidomain profiles, with particularly similar profiles at follow-up. This suggests that, among individuals with persistent symptoms, PCC may increasingly resemble longer-standing ME/CFS across autonomic, small-fiber/sensory, and cognitive domains. These findings are consistent with

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overlapping post-infectious mechanisms, but do not establish identical disease trajectories or definitive disease convergence.

Keywords Autonomic dysfunction, Cognition, Myalgic encephalomyelitis/Chronic fatigue syndrome, Post-COVID condition, Small fiber neuropathy

Introduction

Longitudinal evidence shows a substantial clinical and biological overlap between post-COVID condition (PCC, or long COVID) and Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS), both characterized by long-lasting fatigue, post-exertional malaise, cognitive dysfunction, and markedly reduced quality of life [1–5]. These overlapping post-infectious syndromes are increasingly recognized as part of a shared spectrum of chronic, multisystemic illness involving immune dysregulation, neuroinflammation, and autonomic nervous system (ANS) imbalance [6–12].

Longitudinal studies have demonstrated that approximately 50% of individuals recovering from SARS-CoV-2 infection meet diagnostic criteria for this syndrome for at least 12 months [13, 14]. Fatigue, post-exertional malaise, cognitive impairment, and unrefreshing sleep remain among the most frequent and disabling complaints, with up to 43% of these patients fulfilling ME/CFS diagnostic criteria [1–3]. This symptom overlap complicates differential diagnosis and underscores the need for objective physiological and neuropsychological characterization [1, 15].

The temporal course of both PCC and ME/CFS is typically chronic, fluctuating, and heterogeneous. Although some symptoms, such as fatigue or sleep disturbance, may gradually decline over time, cognitive dysfunction often persists or even becomes more pronounced [16–18]. Most patients experience a relapsing–remitting trajectory rather than sustained recovery, with significant impact on daily functioning, social participation, and employment [1, 19, 20].

Dysautonomia has emerged as a major contributor to the clinical burden in both conditions. Between 27% and 66% of individuals with PCC present moderate to severe autonomic dysfunction, manifesting as orthostatic intolerance, tachycardia, dizziness, or blood pressure instability, often resembling postural orthostatic tachycardia syndrome (POTS) [11, 21–23]. In ME/CFS, chronic autonomic dysfunction affects up to 90% of patients and is frequently characterized by sympathetic predominance and impaired baroreflex control, correlating with greater fatigue severity and cognitive difficulties [16, 24–26]. Neuroimaging and biomarker studies support a neuroimmune basis for these abnormalities, revealing brainstem and white matter changes, endothelial dysfunction, and autoantibodies against adrenergic and muscarinic receptors [9, 27–29].

The long-term impact of these autonomic and cognitive disturbances is profound, often leading to sustained disability, emotional distress, and reduced quality of life [24, 30, 31]. While some longitudinal studies describe partial recovery, a significant subset of patients continues to experience persistent or relapsing symptoms for years [32–35]. Understanding the trajectory and interplay between cognitive and autonomic dysfunction is essential to clarify shared mechanisms and guide clinical management.

In this context, the present study aimed to investigate changes in cognitive, autonomic, and small-fiber abnormalities between two-timepoints in PCC and ME/CFS, and to determine whether these changes differed between diagnostic groups.

Materials and methods

Participants and demographic data

All participants underwent a comprehensive clinical evaluation, including the collection of demographic and clinical data such as age, sex, years of education, disease duration, and other relevant variables. The study cohort comprised 38 individuals, 21 patients diagnosed with PCC and 17 with ME/CFS, who were prospectively evaluated over time for autonomic, cognitive, and small-fiber involvement under standardized diagnostic and ethical procedures.

The inclusion criteria encompassed adults aged between 18 and 70 years with adequate comprehension and communication abilities. Exclusion criteria included pregnancy or lactation, severe trauma, alcoholism, drug addiction, severe cardiac disease, radiological evidence of structural brain pathology, concomitant conditions that could influence study results, and the use of immunomodulatory treatments.

Participants were assigned to the PCC or ME/CFS cohorts according to the primary clinical diagnosis and illness onset recorded at recruitment. PCC participants met the NICE definition for post-COVID condition, with documented or clinically supported SARS-CoV-2 infection preceding symptom onset. Acute infection was confirmed through a positive nasal PCR, the detection of IgG and/or IgM antibodies against SARS-CoV-2, or a medical report supporting the diagnosis. Exclusion criteria specific to this group included respiratory disease lasting beyond 12 weeks, admission to an intensive care unit, or severe bilateral pneumonia requiring hospitalization.

ME/CFS participants had received a clinical diagnosis of ME/CFS according to Fukuda criteria [36] before inclusion in the present study. In this cohort, ME/CFS symptoms preceded any subsequent SARS-CoV-2 infection when present; therefore, participants were not classified as post-COVID-onset ME/CFS. Some ME/CFS participants may have experienced SARS-CoV-2 infection during the follow-up period, potentially worsening pre-existing symptoms, but their primary diagnosis and illness onset were not attributable to COVID-19.

Thus, the two cohorts were not assumed to be mutually exclusive at the symptom-criterion level. Instead, they were defined according to clinical recruitment diagnosis and illness onset: post-COVID-onset chronic illness versus established ME/CFS.

The study protocol was approved by the Basque Drug Research Ethics Committee [*Comité de Ética de la Investigación con medicamentos de Euskadi* (CEIm-E), reference PI2020210], and all participants provided written informed consent in accordance with the Declaration of Helsinki.

Autonomic function assessment

Integrity of the ANS was evaluated through a multimodal battery assessing sudomotor, cardiovascular, and orthostatic domains. Sudomotor function, reflecting sympathetic cholinergic activity, was measured with the Sudoscan® device (Impeto Medica© Paris, France) which quantifies electrochemical skin conductance (ESC, μ S) at the palms and soles. Cardiovascular autonomic control was examined using the Task Force Monitor® (CNSSystems© Graz, Austria) under standardized conditions. Frequency-domain spectral analysis provided heart rate variability (HRV) and blood pressure variability (BPV) indices: high-frequency normalized units (HFnu) were taken as a marker of parasympathetic tone, while low-frequency normalized units (LFnu) of diastolic BP reflected sympathetic vasomotor modulation. Baroreflex sensitivity (BRS) was computed using the sequence method. Cardiovagal (parasympathetic) reactivity was further evaluated through the deep-breathing test. The Valsalva maneuver (40 mmHg for 15 s) enabled the assessment of combined vagal and sympathetic responses, including the Valsalva ratio and systolic pressure recovery time (PRT). Orthostatic responses were assessed via passive head-up tilt at 60° for at least ten minutes with continuous cardiovascular monitoring to detect orthostatic hypotension, POTS, or autonomic failure. All assessments were performed by trained neurologists following standardized protocols to ensure reliability.

Small-fiber function assessment

Thermal and pain small-fiber functions were assessed using the TSA-2 system (Medoc Advanced Medical

System©, Ramat Yishai, Israel), including Contact Heat Evoked Potentials (CHEPs), Cold Evoked Potentials (CEPs), and Quantitative Sensory Testing (QST). CHEPs were obtained by delivering fifteen heat stimuli of 55 °C at 30–45s intervals, while CEPs consisted of fifteen cold stimuli of 9 °C applied at similar intervals. QST was performed to determine thresholds for warm and cold detection as well as for heat and cold pain perception, providing a detailed quantitative evaluation of small sensory fiber integrity.

Neuropsychological and neuropsychiatric evaluation

Neuropsychological and neuropsychiatric assessments were administered by an experienced neuropsychologist. Demographic and clinical variables were recorded for all participants, and overall cognitive function was screened using the Montreal Cognitive Assessment (MoCA) [37, 38]. A comprehensive cognitive battery was subsequently performed to evaluate attention, executive function, memory, language, and perceptual abilities. Attention and processing speed were assessed with the Trail Making Test A (TMT-A) [39] and the Salthouse Perceptual Comparison Test (SPCT) [40, 41]; executive function and cognitive flexibility were measured through the Trail Making Test B (TMT-B), the Modified Wisconsin Card Sorting Test (M-WCST) [42, 43], and the Stroop Test [44]; verbal fluency was evaluated using both semantic (animals) and phonemic (P) categories; memory was examined with the Hopkins Verbal Learning Test–Revised (HVLT-R) [45]; and perceptual abilities were assessed with the Benton Judgment of Line Orientation (JLO) [46]. Sustained attention was evaluated using the Toulouse-Piéron Revised (TP-R) test [47]. Neuropsychiatric status was further characterized with validated questionnaires addressing fatigue (Modified Fatigue Impact Scale, MFIS) [48], depressive symptoms (Geriatric Depression Scale, GDS) [49], anxiety (State-Trait Anxiety Inventory, STAI) [50], suicidal ideation (Columbia Suicide Severity Rating Scale, C-SSRS) [51], and sleep quality (Pittsburgh Sleep Quality Index, PSQI) [52].

Statistical analysis

Statistical analyses were performed using IBM SPSS Statistics 26.0 and R (version 4.4.2). Descriptive statistics were used to summarize demographic and clinical characteristics, expressed as median and interquartile range (IQR) for continuous variables and counts and percentages for categorical variables. Because several variables showed non-normal distributions and the sample size was relatively small, non-parametric approaches were applied.

Eighteen domain-level composite scores were created to summarize the multidomain assessment and reduce the number of individual outcome measures. The

domains included Autonomic Symptoms, Hemodynamic function, Sympathetic function, Parasympathetic function, Small Fiber Symptoms, CHEPs, CEPs, QST Warm, QST Cold, Fatigue, Psychiatric symptoms, Fluency, Processing Speed, Attention, Verbal Memory, Executive Function, Visuospatial function, and Global Cognition. Analyses were conducted at the domain level; therefore, the multidomain approach refers to the parallel assessment of complementary autonomic, small-fiber/sensory, cognitive, and clinical symptom domains rather than formal multimodal statistical integration.

Cross-domain associations at baseline were assessed using Spearman rank correlation coefficients, and within-subject change from baseline to follow-up using Wilcoxon signed-rank tests. Repeated-measures effects were examined using rank-transformed linear mixed-effects models fitted separately for each domain, with subject-level random intercepts and Time (baseline vs. follow-up) as the primary fixed effect. Additional models tested Time $\times$ Group and Time $\times$ disease duration effects. To assess the potential influence of baseline disease-duration imbalance, sensitivity models testing Time $\times$ Group were repeated including baseline disease duration and the Time $\times$ disease duration interaction. In addition, complete covariate-adjusted rank-transformed mixed-effects models were fitted across the 18 domains, including Time, Group, Time $\times$ Group, baseline disease duration, Time $\times$ disease duration, age, education level, interval between assessments, and subject-level random intercepts. Full model outputs are reported in Supplementary Tables 1–5.

To examine whether baseline domain status predicted two-timepoint follow-up change, median quantile regression models were fitted with baseline values as predictors and longitudinal change (follow-up minus baseline) as the outcome. This analysis was used to explore baseline-dependent effects in domains showing FDR-significant Time effects in the fully adjusted models.

A two-tailed significance threshold of $p < 0.05$ was applied throughout, and false discovery rate (FDR) correction was applied within analytic families. FDR correction was performed using the Benjamini–Hochberg procedure, applied separately within each family of effects across the 18 prespecified domains. FDR-adjusted p -values ($p\text{FDR}$) < 0.05 were considered statistically significant.

Results

Clinical and demographic characteristics

The study included 38 participants, 21 with PCC and 17 with ME/CFS. The PCC group comprised 17 women (81%), with a median age of 51 years (IQR = 47.5–57.0), and the ME/CFS group included 16 women (94.1%), with a median age of 43 years (IQR = 36.5–50.0).

Median disease duration at baseline was 12 months (IQR = 10.5–24.5) for PCC and 42 months (IQR = 32–100) for ME/CFS. The median interval between evaluations was comparable (PCC: 30 months, IQR = 24.5–36.5; ME/CFS: 31 months, IQR = 26–38.5). Educational level did not differ significantly between groups (median 15 years in both).

Baseline domain-level comparisons

At baseline, ME/CFS and PCC showed largely overlapping profiles across sensory, autonomic, cognitive, and clinical symptom domains (Table 1). However, nominal between-group differences emerged in some physiological measures. ME/CFS patients exhibited higher QST warm detection thresholds compared with PCC (median [IQR]: 0.62 [–0.25–0.81] vs. –0.11 [–0.72–0.52]; $p = 0.034$), indicating greater thermal sensory dysfunction. In the autonomic domain, ME/CFS showed a higher burden of autonomic symptoms as assessed by COM-PASS-31 (0.60 [–0.33–1.10] vs. –0.27 [–1.09–0.21]; $p = 0.038$), alongside lower hemodynamic composite scores compared with PCC (–0.14 [–0.35–0.02] vs. 0.18 [–0.25–0.53]; $p = 0.019$). None of these differences remained significant after FDR correction. No significant between-group differences were observed across cognitive domains or clinical symptom measures at baseline.

Multidimensional correlation patterns

Significant cross-domain associations emerged between cognitive, sensory, autonomic, and symptom domains at baseline (Fig. 1). Cognitive domains showed strong positive intercorrelations, indicating internal consistency across neuropsychological composite scores. Markers of peripheral sensory and autonomic dysfunction were significantly related to symptom burden. Small fiber symptom severity was positively associated with autonomic symptom burden ($\text{Rho} = 0.65$, $p\text{FDR} = 0.002$) and with fatigue severity ($\text{Rho} = 0.55$, $p\text{FDR} = 0.018$). Fatigue showed consistent negative associations with multiple cognitive domains, including processing speed ($\text{Rho} = -0.57$, $p\text{FDR} = 0.004$), attention ($\text{Rho} = -0.49$, $p\text{FDR} = 0.018$), and executive function ($\text{Rho} = -0.44$, $p\text{FDR} = 0.047$), suggesting that greater fatigue burden was associated with lower cognitive performance.

A moderate association was also observed between autonomic symptom burden and fatigue ($\text{Rho} = 0.47$, $p = 0.007$, $p\text{FDR} = 0.050$), although this result was borderline after FDR correction. Associations involving psychiatric symptoms and evoked potential measures were weaker and did not survive FDR correction.

Longitudinal changes from baseline to follow-up

At follow-up, no significant differences between-group were observed across sensory/small-fiber, autonomic,

Table 1 Baseline domain-level z-scores in patients with ME/CFS and PCC

	ME/CFS Median (IQR)	PCC Median (IQR)	Statistics (Wilcoxon)
Sensory / Small-fiber function			
CHEPs	−0.30 (−0.78–0.11)	−0.26 (−0.44–0.14)	W = 174
CEPs	0.09 (−0.15–0.30)	0.15 (−0.03–0.46)	W = 190
QST Cold	0.30 (−0.24–0.69)	0.46 (0.08–0.78)	W = 195
QST Warm	0.62 (−0.25–0.81)	−0.11 (−0.72–0.52)	W = 100 *
SFNSL	−0.46 (−1.14–0.88)	−0.32 (−0.85–0.29)	W = 95
Autonomic function			
COMPASS-31	0.60 (−0.33–1.10)	−0.27 (−1.09–0.21)	W = 78 *
Sympathetic function	0.10 (−0.04–0.54)	0.08 (−0.14–0.55)	W = 131
Parasympathetic function	−0.06 (−0.61–0.34)	−0.23 (−0.86–0.70)	W = 116
Hemodynamic function	−0.14 (−0.35–0.02)	0.18 (−0.25–0.53)	W = 245 *
Cognitive domains			
Global Cognition	0.10 (−0.36–0.26)	−0.00 (−0.23–0.13)	W = 146
Verbal Memory	0.02 (−0.53–0.47)	0.09 (−0.75–0.29)	W = 166
Executive Function	0.08 (−0.29–0.33)	0.13 (−0.08–0.26)	W = 181
Processing Speed	−0.04 (−0.49–0.45)	0.00 (−0.60–0.55)	W = 159.5
Attention	−0.03 (−0.17–0.10)	0.04 (−0.24–0.17)	W = 170
Verbal Fluency	0.23 (−0.35–0.71)	0.26 (−0.38–0.62)	W = 151
Visuospatial function	0.18 (−0.74–0.24)	−0.27 (−0.69–0.75)	W = 175
Clinical symptom domains			
Fatigue	0.37 (−0.48–0.76)	0.22 (−0.24–0.82)	W = 159
Psychiatric symptoms	0.19 (−0.46–0.51)	−0.01 (−0.15–0.46)	W = 147

* $p \leq 0.05$. Values are presented as median (interquartile range, IQR). Between-group comparisons were performed using the Wilcoxon rank-sum test, and the corresponding W statistic is reported. P-values are uncorrected for multiple comparisons; none of the comparisons remained statistically significant after false discovery rate (FDR) correction. Abbreviations: CHEPs, contact heat-evoked potentials; CEPs, cold evoked potentials; QST, quantitative sensory testing (cold and warm detection thresholds); SFNSL, Small Fiber Neuropathy Symptom List; COMPASS-31, Composite Autonomic Symptom Score-31; Sympathetic function, composite z-score of sympathetic autonomic indices; Parasympathetic function, composite z-score of parasympathetic autonomic indices; Hemodynamic function, composite z-score of cardiovascular hemodynamic measures; Global Cognition, composite z-score across cognitive domains; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; PCC, post-COVID condition

cognitive, or clinical symptom domains (Table 2). All domain-level comparisons showed overlapping distributions, with comparable median z-scores and interquartile ranges between groups, and none of the Wilcoxon rank-sum tests reached statistical significance. These findings indicate a convergence of physiological and clinical profiles between ME/CFS and PCC at follow-up.

Paired within-subject analyses using Wilcoxon signed-rank tests revealed several nominal changes from baseline to follow-up across autonomic, small-fiber/sensory, and cognitive domains (Fig. 2). In the autonomic domain, parasympathetic response increased (median change = 0.31 SD; $p = 0.027$), while sympathetic activity decreased (median change = −0.43 SD; $p = 0.040$), suggesting partial autonomic rebalancing. In the small-fiber/sensory domain, CHEPs showed a marked increase from baseline to follow-up (median change = 0.63 SD; $p < 0.001$), consistent with an apparent improvement in small-fiber nociceptive responses, and small-fiber symptom burden improved (median change = 0.38 SD; $p = 0.048$). Sensory cold thresholds decreased over time (median change = −0.18 SD; $p = 0.009$), indicating worsening in the QST cold domain. In the cognitive

domain, verbal memory increased significantly (median change = 0.28 SD; $p = 0.006$).

Rank-transformed linear mixed-effects models were then used to evaluate repeated-measures effects across domains (Supp. Tables 1–4). In the primary Time-only models, FDR-corrected Time effects were observed in the small-fiber/sensory domain for CHEPs ($pFDR < 0.001$) and QST cold thresholds ($pFDR = 0.038$), and in the cognitive domain for processing speed ($pFDR = 0.006$) and verbal memory ($pFDR = 0.010$). Unadjusted Time $\times$ Group models suggested differential follow-up changes in autonomic-related domains, specifically hemodynamic function and autonomic symptom burden (both $pFDR = 0.027$). However, these effects did not remain significant after adjustment for baseline disease duration (Supp Table 4). Similarly, disease duration did not consistently modulate follow-up changes: no Time $\times$ Disease duration interaction survived FDR correction.

Complementary complete covariate-adjusted rank-transformed mixed-effects models were fitted across all 18 domains, including Time, Group, Time $\times$ Group, baseline disease duration, Time $\times$ Disease duration, age, education level, interval between assessments, and subject-specific random intercepts (Supp Table 5). In these

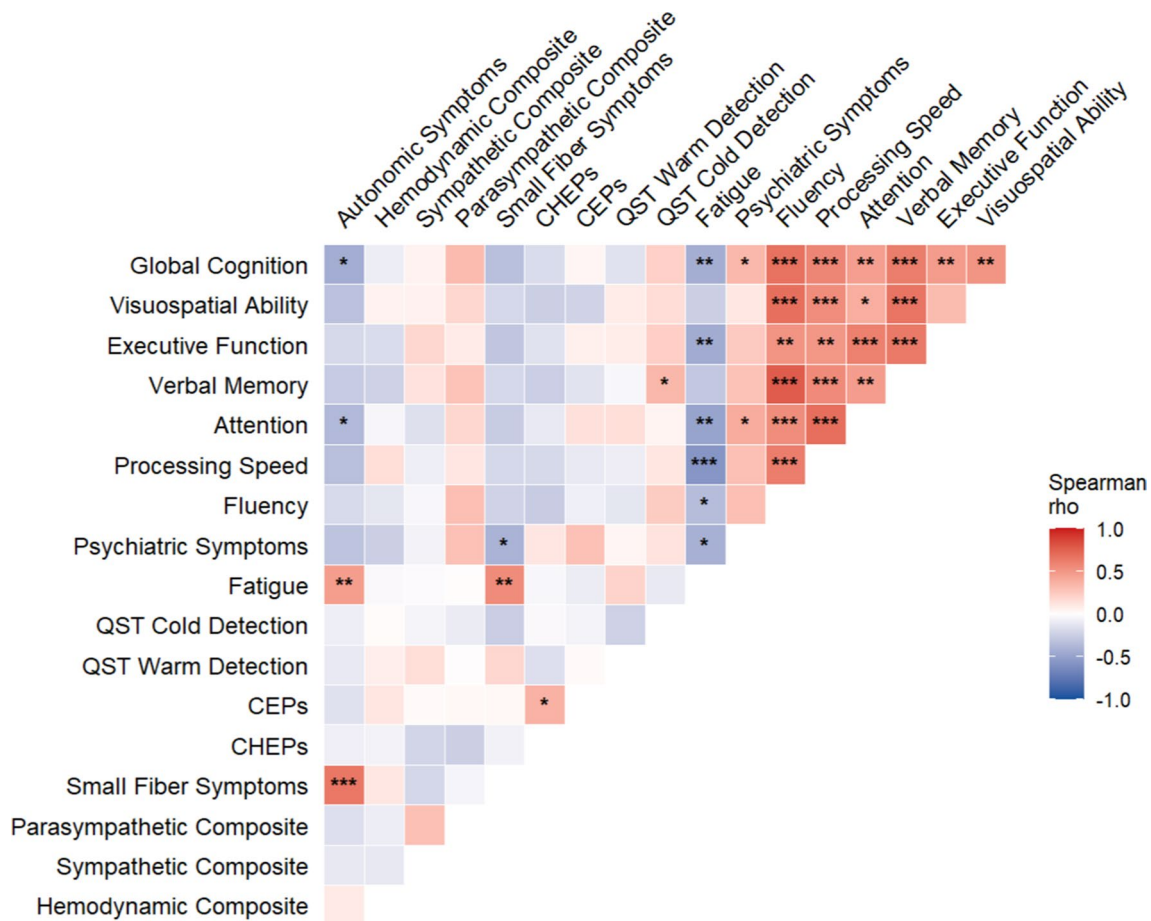


Fig. 1 Baseline cross-domain associations between domain-level composite scores. The heatmap shows Spearman correlation coefficients between autonomic, small-fiber/sensory, cognitive, and clinical symptom domains at baseline. Color intensity represents Spearman's rho. Asterisks indicate uncorrected significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. FDR-corrected associations are reported in the text. CEPs: cold evoked potentials; CHEPs: contact heat-evoked potentials; COMPASS-31: Composite Autonomic Symptom Score-31; QST: quantitative sensory testing

models, FDR-corrected Time effects remained significant for CHEPs ($\beta = 22.19$, $p\text{FDR} < 0.001$), verbal memory ($\beta = 12.30$, $p\text{FDR} = 0.004$), processing speed ($\beta = -9.42$, $p\text{FDR} = 0.005$), and QST cold thresholds ($\beta = -9.32$, $p\text{FDR} = 0.044$) (Fig. 3). No Group, Time $\times$ Group, disease duration, Time $\times$ Disease duration, education level, or interval between assessments effect survived FDR correction. Age was significantly associated with parasympathetic function ($\beta = -1.10$, $p\text{FDR} = 0.003$).

Baseline-dependent change analyses were restricted to domains showing FDR-significant Time effects in the complete covariate-adjusted models (Supp Fig. 1). In median quantile regression models, lower baseline values were associated with greater follow-up change in CHEPs responses ($\beta = -1.19$, $p\text{FDR} < 0.001$) and processing speed ($\beta = -0.32$, $p\text{FDR} = 0.003$), indicating an inverse association between baseline status and subsequent change. By contrast, baseline values were not significantly associated with follow-up change in QST cold detection/pain or verbal memory after FDR correction.

Discussion

Building on the increasing recognition that PCC and ME/CFS may represent points along a shared post-infectious spectrum, the present study was designed to characterize their physiological changes from baseline to follow-up using a comprehensive multidomain assessment. By assessing autonomic function, small-fiber integrity, clinical symptoms, and neurocognitive performance at two time points, we were able to examine both within-subject change and cross-domain relationships across two assessment timepoints.

Within this framework, our findings provide evidence of substantial overlap between PCC and ME/CFS across autonomic, sensory, and cognitive domains at baseline, with selective differences restricted to sensory and autonomic measures. ME/CFS patients exhibited greater thermal sensory dysfunction and higher autonomic symptom burden, likely reflecting longer disease duration rather than distinct mechanisms. Cognitive performance did not differ between groups, arguing against

Table 2 Follow-up domain-level z-scores in patients with ME/CFS and PCC

	ME/CFS Median (IQR)	PCC Median (IQR)	Statistics (Wilcoxon)
Sensory / Small-fiber function			
CHEPs	0.41 (0.07–0.51)	0.32 (0.20–0.41)	W = 135
CEPs	0.17 (–0.07–0.40)	–0.05 (–0.45–0.24)	W = 101
QST Cold	–0.14 (–0.63–0.77)	–0.40 (–0.73–0.36)	W = 131
QST Warm	0.14 (–0.56–0.98)	–0.11 (–0.74–0.34)	W = 120
SFNSL	0.20 (–0.73–1.11)	0.17 (–0.39–0.85)	W = 154
Autonomic function			
COMPASS-31	0.38 (–0.57–0.58)	–0.01 (–0.70–0.71)	W = 113
Sympathetic function	0.11 (–0.45–0.20)	–0.12 (–0.37–0.09)	W = 63
Parasympathetic function	0.47 (–0.37–1.37)	–0.13 (–0.44–0.36)	W = 95
Hemodynamic function	0.11 (–0.20–0.32)	–0.05 (–0.33–0.15)	W = 143
Cognitive domains			
Global Cognition	0.16 (–0.10–0.47)	–0.09 (–0.34–0.15)	W = 92
Verbal Memory	0.51 (–0.04–0.87)	–0.08 (–0.57–0.64)	W = 141
Executive Function	0.09 (–0.23–0.30)	0.01 (–0.41–0.19)	W = 127
Processing Speed	0.11 (–0.28–0.38)	–0.18 (–0.79–0.29)	W = 136
Attention	–0.04 (–0.18–0.10)	–0.03 (–0.30–0.22)	W = 144
Verbal Fluency	0.31 (–0.48–0.77)	0.00 (–0.81–0.60)	W = 147
Visuospatial function	–0.10 (–0.95–0.50)	–0.27 (–0.86–0.24)	W = 172
Clinical symptom domains			
Fatigue	0.34 (–1.35–0.67)	0.46 (–0.39–0.76)	W = 185
Psychiatric symptoms	0.33 (–0.91–0.63)	0.02 (–0.52–0.59)	W = 174

Values are presented as median (interquartile range, IQR). Between-group comparisons were performed using the Wilcoxon rank-sum test, and the corresponding W statistic is reported. P-values are uncorrected for multiple comparisons; none of the comparisons remained statistically significant after false discovery rate (FDR) correction. Abbreviations: CHEPs, contact heat-evoked potentials; CEPs, cold evoked potentials; QST, quantitative sensory testing (cold and warm detection thresholds); SFNSL, Small Fiber Neuropathy Symptom List; COMPASS-31, Composite Autonomic Symptom Score-31; Sympathetic function, composite z-score of sympathetic autonomic indices; Parasympathetic function, composite z-score of parasympathetic autonomic indices; Hemodynamic function, composite z-score of cardiovascular hemodynamic measures; Global Cognition, composite z-score across cognitive domains; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; PCC, post-COVID condition

duration-dependent cognitive decline as a defining feature of either condition. These results are consistent with the possibility of overlapping post-infectious mechanisms, consistent with previous reports [10, 53–55].

From baseline to follow-up, both groups showed broadly overlapping patterns of change across most domains. The most robust change was a significant increase in nociceptive small-fiber function, reflected by shorter CHEPs latencies. However, this CHEPs finding requires particular caution. The strong inverse association between baseline CHEPs values and subsequent change is compatible with regression to the mean, because participants with the most impaired baseline responses showed the largest increases over time. Therefore, the observed CHEPs increase may reflect true physiological recovery, baseline-dependent plasticity, measurement variability, statistical regression, or a combination of these factors. In contrast, cold sensory thresholds worsened modestly, indicating modality-specific and incomplete sensory change, consistent with previous longitudinal small fiber neuropathy observations [56].

Autonomic changes were characterized by increased parasympathetic and reduced sympathetic activity over

time, suggesting partial autonomic rebalancing rather than global recovery. Hemodynamic measures remained largely stable, consistent with cardiovascular deconditioning [57]. Clinically overt orthostatic syndromes were uncommon, supporting the heterogeneous nature of autonomic dysfunction in PCC and ME/CFS [11]. Unadjusted models suggested possible Time × Group effects in hemodynamic function and autonomic symptom burden, but these effects did not remain significant after adjustment for baseline disease duration. Therefore, evidence for group-specific autonomic change should be interpreted cautiously.

Cognitive changes were modest and selective. Verbal memory showed a small but significant improvement over time, whereas processing speed showed a significant decrease in the rank-based longitudinal models, while global cognition, executive function, attention, and verbal fluency remained largely stable [58, 59]. The improvement in verbal memory should also be interpreted cautiously, as repeated neuropsychological testing may introduce practice effects. Conversely, changes in processing speed may reflect true cognitive change, fluctuation in fatigue or attentional resources, or test-retest

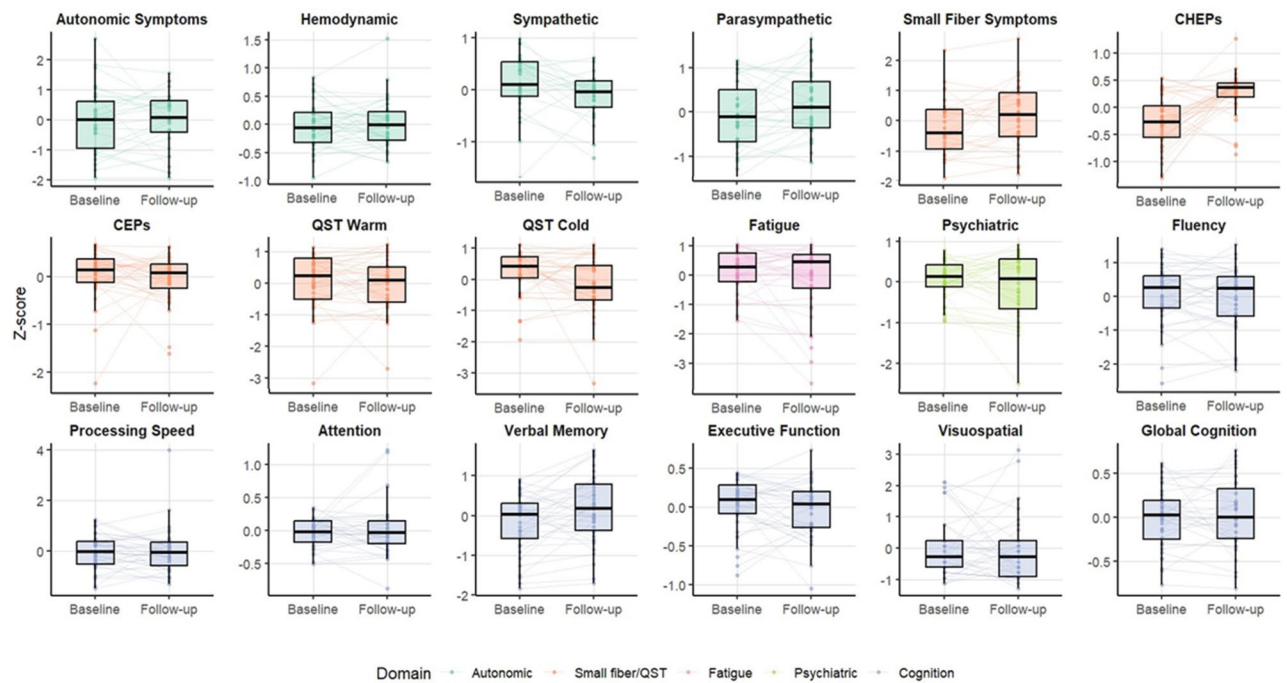


Fig. 2 Baseline and follow-up changes across autonomic, sensory, fatigue, psychiatric, and cognitive domains. Changes were assessed using Wilcoxon signed-rank tests. Values represent median change in domain-level z-scores from baseline to follow-up. Paired boxplots and individual trajectories illustrating within-subject change from baseline to follow-up across 18 physiological and cognitive domains. Each point represents a participant's z-score (normalized to baseline variance), lines indicate individual change, and boxes show group distributions. Domains are color-coded by system. Higher scores reflect better performance or lower physiological burden in all domains except Fatigue and Psychiatric symptoms, where higher values represent greater severity (worse clinical status)

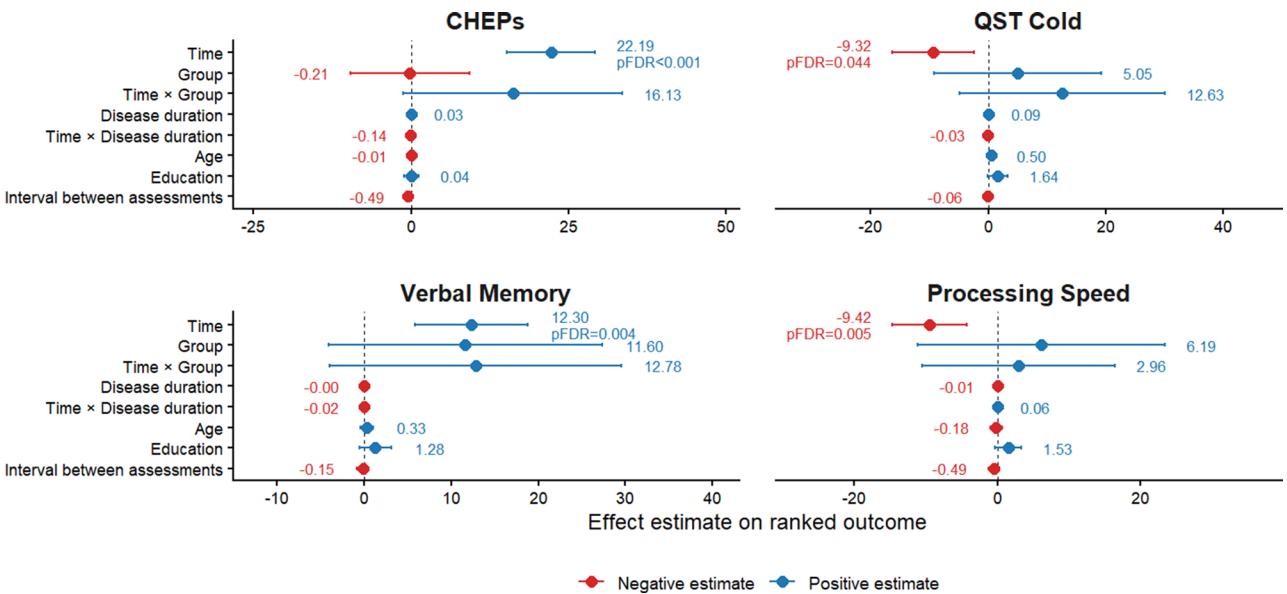


Fig. 3 Covariate-adjusted longitudinal models in domains showing significant global Time effects. Forest plots show fixed-effect estimates and 95% confidence intervals from rank-transformed linear mixed-effects models for domains showing significant FDR-corrected longitudinal Time effects in the primary analysis. Models included Time, Group, Time × Group, age, education level, disease duration, interval between assessments, and subject-specific random intercepts. Positive estimates indicate higher ranked scores, whereas negative estimates indicate lower ranked scores. pFDR values displayed for the Time effect correspond to the covariate-adjusted models shown in the figure

variability. The absence of widespread cognitive deterioration argues against progressive neurodegeneration and supports a model of relative cognitive stability with limited domain-specific change. Importantly, no cognitive domain exhibited robust FDR-corrected evidence of differential change between PCC and ME/CFS after adjustment for disease duration, further reinforcing broadly overlapping patterns of change.

Despite objective changes in several physiological domains, subjective symptom measures showed limited parallel improvement. This dissociation between physiological markers and perceived symptom burden likely reflects complex interactions between chronic illness, central sensory processing, fatigue, and illness perception. Persistent symptom severity in the context of partial physiological change has been reported in other chronic post-infectious conditions and may be reinforced by long-standing maladaptive fatigue–pain–sleep cycles and heightened interoceptive awareness [60].

From a clinical perspective, these findings highlight the value of comprehensive multidomain assessment in PCC and ME/CFS. The observed domain-specific changes suggest that targeted interventions such as autonomic rehabilitation and symptom-oriented management may leverage residual physiological reserve. At the same time, the persistence of subjective burden despite objective changes underscores the need for approaches that address central symptom integration, illness perception, and long-term coping strategies [60, 61].

Several limitations warrant consideration. The modest sample size, particularly given the multidomain nature of the study and the number of outcomes assessed, limits its statistical power to detect subtle group-specific effects and increases the risk of sample-specific findings. The two-timepoint follow-up design also precludes characterization of dynamic trajectories, and the absence of a healthy longitudinal control group limits the ability to distinguish true biological change from compensatory adaptation, test-retest variability, practice effects, or regression to the mean. Baseline disease duration also differed substantially between groups and may contribute to variability in follow-up change. Although duration-adjusted sensitivity analyses did not show FDR-significant Time $\times$ Group effects or consistent modulation of change by disease duration, residual confounding cannot be excluded. Accordingly, the absence of robust group-specific effects should not be interpreted as definitive evidence of disease convergence or identical follow-up patterns. Rather, these findings support broadly overlapping multidomain profiles within the limits of the available sample and should be replicated in larger cohorts with additional follow-up timepoints and appropriate control groups.

Conclusions

In summary, this two-timepoint follow-up study supports substantial overlap between PCC and ME/CFS across autonomic, small-fiber/sensory, cognitive, and clinical symptom domains. Among individuals with persistent symptoms, PCC may come to resemble longer-standing ME/CFS across several physiological and cognitive domains.

Abbreviations

ANS	Autonomic nervous system
BPV	Blood pressure variability
BRS	Baroreflex sensitivity
CEPs	Cold Evoked Potentials
CHEPs	Contact Heat Evoked Potentials
C-SSRS	Columbia Suicide Severity Rating Scale
ESC	Electrochemical skin conductance
GDS	Geriatric Depression Scale
HFnu	High-frequency normalized units
HRV	Heart rate variability
HVLT-R	Hopkins Verbal Learning Test–Revised
IQR	Interquartile range
JLO	Judgment of Line Orientation
LFnu	Low-frequency normalized units
ME/CFS	Myalgic encephalomyelitis/chronic fatigue syndrome
MFIS	Modified Fatigue Impact Scale
MoCA	Montreal Cognitive Assessment
M-WCST	Modified Wisconsin Card Sorting Test
PCC	Post-COVID condition
POTS	Postural orthostatic tachycardia syndrome
PRT	Pressure recovery time
PSQI	Pittsburgh Sleep Quality Index
QST	Quantitative Sensory Testing
SPCT	Salthouse Perceptual Comparison Test
STAI	State-Trait Anxiety Inventory
TMT	Trail Making Test
TP-R	Toulouse–Piéron Revised

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-026-08321-9>.

Supplementary Material 1

Authors' contributions

NA: conceptualization, data curation, formal analysis, investigation, visualization, writing – original draft. CB, BT-M, MA, TF-V, JVL, IG, MR-L: investigation, data acquisition, and critical revision of the manuscript. JVL: funding acquisition. RDP, BT-M, and JCGE: conceptualization, supervision, funding acquisition, and writing – review & editing. All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Basque Drug Research Ethics Committee (Comité de Ética de la Investigación con Medicamentos de Euskadi, CEIm-E; reference PI2020210). All participants provided written informed consent prior to inclusion in the study, in accordance with the Declaration of Helsinki.

Competing interests

The authors declare that they have no competing interests.

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